



CHAPTER

CIP Tank

The concentrate tank, in CIP mode: what's really inside it, and why that matters

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17.1 Overview and function

No HydroVolta document identifies a separate, dedicated CIP tank — **the concentrate tank (chapter 16) is the one and only vessel used for CIP** cleaning, per HydroVolta's own documented CIP procedure (chapter 12). It gets its own chapter here because what is *inside* that tank changes completely depending on which mode the unit is in, and confusing the two is a real chemical-exposure hazard — read this chapter alongside chapters 12 and 16, not instead of them.



Figure 17.1. Suggested figure: the same concentrate tank as fig. 16.1, with the CIP dosing/circulation connections called out separately from the normal process connections.

17.2 Key specifications: what is in the tank, and when

- **Production mode:** the tank holds **concentrate/ brine** — rejected feedwater ions, increasingly concentrated over multiple passes (chapter 16). This is process water, not a cleaning chemical.
- **CIP acid step:** the tank holds **pretreated feedwater dosed with HCl to pH 2.5** (chapter 12) — an acidic solution, corrosive at that pH.
- **Between CIP steps:** the tank is drained and refushed with **fresh feed water** for 15–20 minutes specifically to clear the acid before the next chemical goes in.
- **CIP hypochlorite step:** the tank holds **feedwater dosed with NaOCl to 2 ppm** (sodium hypochlorite, effective only near pH 7–7.5) — an oxidizing, corrosive solution.
- **NaOH variant of CIP** (electrolyte-compartment cleaning): the tank holds **NaOH dissolved only in pure (permeate-quality) water** — **never** feedwater or process water, per the warning in chapter 12 (mixing NaOH with the minerals present in feed/ process water causes scaling).

No capacity/material specification exists yet for a finalized standard SalinBloc™ bill of materials — see chapters 15 and 16 for the real tank orders on file and their (unconfirmed) role.

17.3 Operation



DANGER

Never let the acid (HCl) and hypochlorite (NaOCl) steps mix in this tank. Mixing an acid with sodium hypochlorite releases chlorine gas, which is toxic even in small quantities. This is precisely why HydroVolta's documented procedure requires a full drain and a 15–20 minute fresh-water flush between the acid step and the hypochlorite step (chapter 12) — never shorten or skip that flush to save time, and never dose both chemicals into the tank at once. If chlorine odor is detected during or after a CIP cycle, stop dosing immediately, ventilate the area, and do not re-enter until it has cleared — see section 2.5 for first-aid measures for chemical/inhalation exposure.

Both the acid and hypochlorite solutions are corrosive at the concentrations used (see the Safety Data Sheet summary in section 2.5): treat the tank, its vent, and any sample point as a chemical-exposure risk during CIP, not just during dosing-pump refilling. Wear the PPE specified in section 2.3 whenever the tank is opened, sampled, or drained during or immediately after a CIP cycle. Do not start a CIP cycle without first confirming the tank's level and that the CIP injection nozzle is submerged (chapter 12).

17.4 Maintenance: cleaning the tank itself

The documented CIP procedure already *is* the tank-cleaning procedure — there is no additional physical scrub-down step on file. In order: drain the acid step, flush with fresh feed water 15–20 minutes; drain the hypochlorite step 5 minutes, then flush with fresh feed water 30 minutes (chapter 12). TODO: no document describes a physical inspection (scale/residue check on the tank walls, fittings, level sensor) as part of or after this cycle — until one exists, a visual check after the final flush (before returning to Production mode) is good practice. See chapter 16 for the tank's general (non-CIP) maintenance notes.

17.5 Troubleshooting

If a CIP cycle fails to reach its target pH (2.5 for the acid step), verify the tank's level and injection nozzle before suspecting a dosing- pump fault (chapter 12). If chlorine odor is ever detected, treat it as an emergency per the warning above, not a troubleshooting item to work through calmly. `HL_Alarm_TC/LL_Alarm_TC` (chapter 16) apply during CIP exactly as they do during Production — do not disable or ignore them because the tank is in CIP mode.